Scenario 1: Forecast for Next Week

You want to predict the next 7 days of website traffic. What should you do?

Step-by-step Solution:

1. Check trend/seasonality:
   If there is a pattern → Use Holt-Winters or SARIMA.
   If not → Use simple ARIMA or Moving Average.
2. Train on past data.
3. Forecast using model:
   Use $.forecast(7)$ to get next week's values.
4. Plot to compare.
   Conclusion: Use a model based on the pattern and forecast ahead.

Scenario 2: Choosing a Time Series Model

Your data has no trend and no seasonality. What model will you choose?

Step-by-step Solution:

1. Plot data → Flat, no clear pattern.
2. No trend or seasonality? → Use simple models like:
   AR (AutoRegressive)
   Moving Average (MA)
3. Fit and test model.
   Conclusion: For flat series → AR or MA model is enough.

Scenario 3: Monthly Sales Data Showing Steady Growth

You observe that monthly sales data for a company has been increasing steadily over the last 2 years. How would you forecast future sales?

Logical Solution:

1. First, visualize the sales trend to confirm consistent growth.
2. Recognize that this indicates a **trend** component in the time series.
3. Check if the series is stationary — a stationary series has no trend or seasonality.
4. Since it has a trend, apply **differencing** to make the data stationary.
5. Select a suitable forecasting model that handles trend, like **ARIMA**.
6. Train the model on past data and generate forecasts for future months.
   Conclusion: Since the data has a trend and no seasonality, use ARIMA after making it stationary.

Scenario 4: Periodic Increase in Sales Every December
Sales data for a retail store shows a spike every December. How do you handle this while forecasting?

Logical Solution:

1. Identify the repeating pattern — this is **seasonality**.
2. Confirm this by visualizing multiple years of data.
3. Use a model that can explicitly account for seasonality such as **Holt-Winters** or **SARIMA**.
4. Fit the seasonal model on historical data.
5. Use the model to forecast future values, ensuring it captures December peaks.
   Conclusion: Seasonality requires models that understand repeated patterns — SARIMA or Holt-Winters.

Scenario 5: Handling Missing Values in Time Series

You are analyzing a time series with missing daily values. What approach will you take?

Logical Solution:

1. Assess how much and where the data is missing — randomly or in blocks.
2. If only a few values are missing, they can be filled using values from nearby dates (interpolation or previous values).
3. If many values are missing or at the start/end, more caution is needed.
4. Never ignore or drop time points casually in a time series as time order matters.
5. After handling missing data, proceed with model building.
   Conclusion: Fill missing values in a way that preserves time continuity and data integrity.

Scenario 6: Evaluating the Accuracy of Forecasts

You've built a time series model. How will you evaluate if it's accurate?

Logical Solution:

1. Divide your data into training and test sets.
2. Train your model on the training portion and generate forecasts.
3. Compare forecasts with actual test data using metrics like:
   Mean Absolute Error (MAE)
   Root Mean Squared Error (RMSE)
   Mean Absolute Percentage Error (MAPE)
4. The smaller the error values, the better the model.
5. Compare different models based on these metrics to select the best one.
   Conclusion: Always validate time series models using real data and appropriate error metrics.